GEOGRAPHY

PHYSICAL GEOGRAPHY

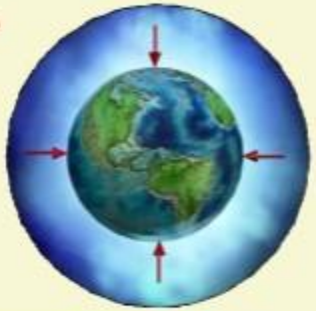
CLIMATOLOGY

MODULE – 16

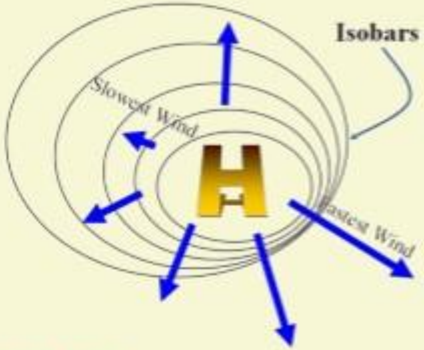
ATMOSPHERIC PRESSURE

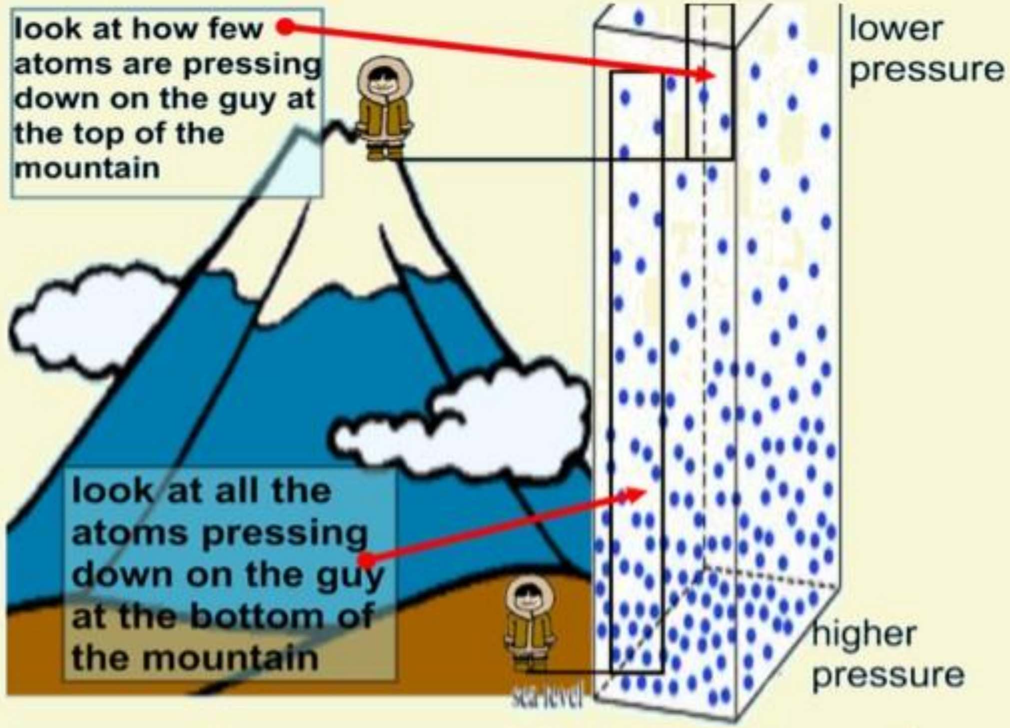
CLIMATOLOGY MODULES

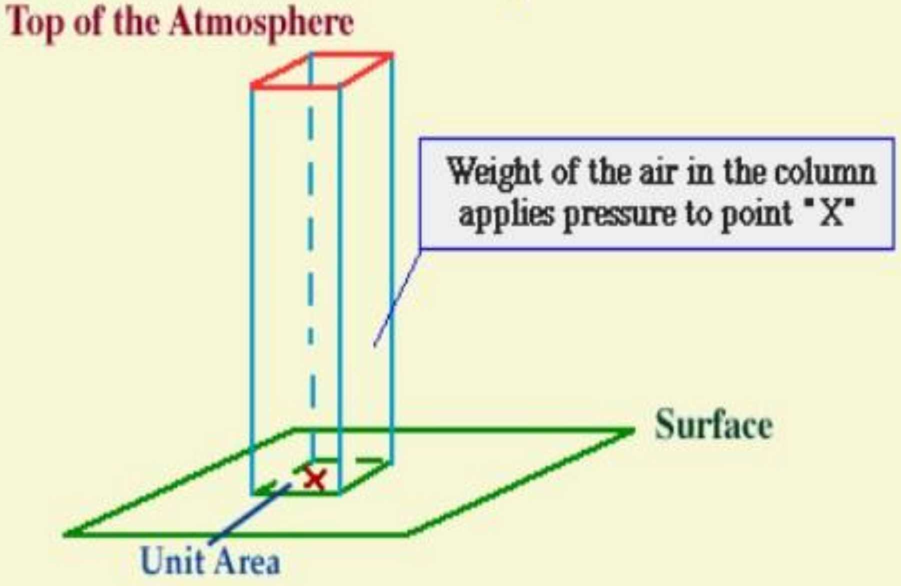
Atmosphere	▪ Composition and structure; weather and climate;
Insolation	▪ Factors controlling insolation; heat budget of the earth – heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection);
Temperature	▪ Factors controlling temperature; distribution of temperature – horizontal and vertical; inversion of temperature;
Pressure	▪ Pressure belts; winds-planetary, seasonal and local; air masses and fronts; tropical and extra-tropical cyclones;
Precipitation	▪ Evaporation; condensation – dew, frost, fog, mist and cloud; rainfall – types and world distribution;
World Climate and Climatic Changes	▪ Different types of Climate



ATMOSPHERIC PRESSURE

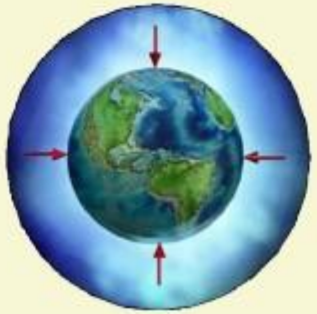






1 The weight of a column of air contained in a unit area from the mean sea level to the top of the atmosphere is called the atmospheric pressure.

2 Pressure is measured with the help of aneroid or mercury barometer and it is expressed in units of millibar.



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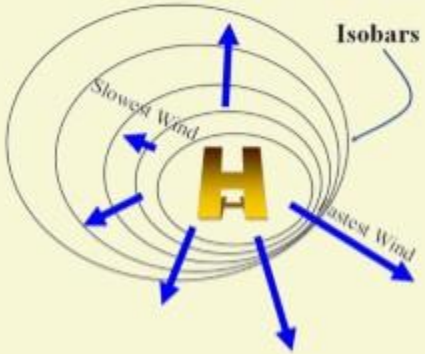
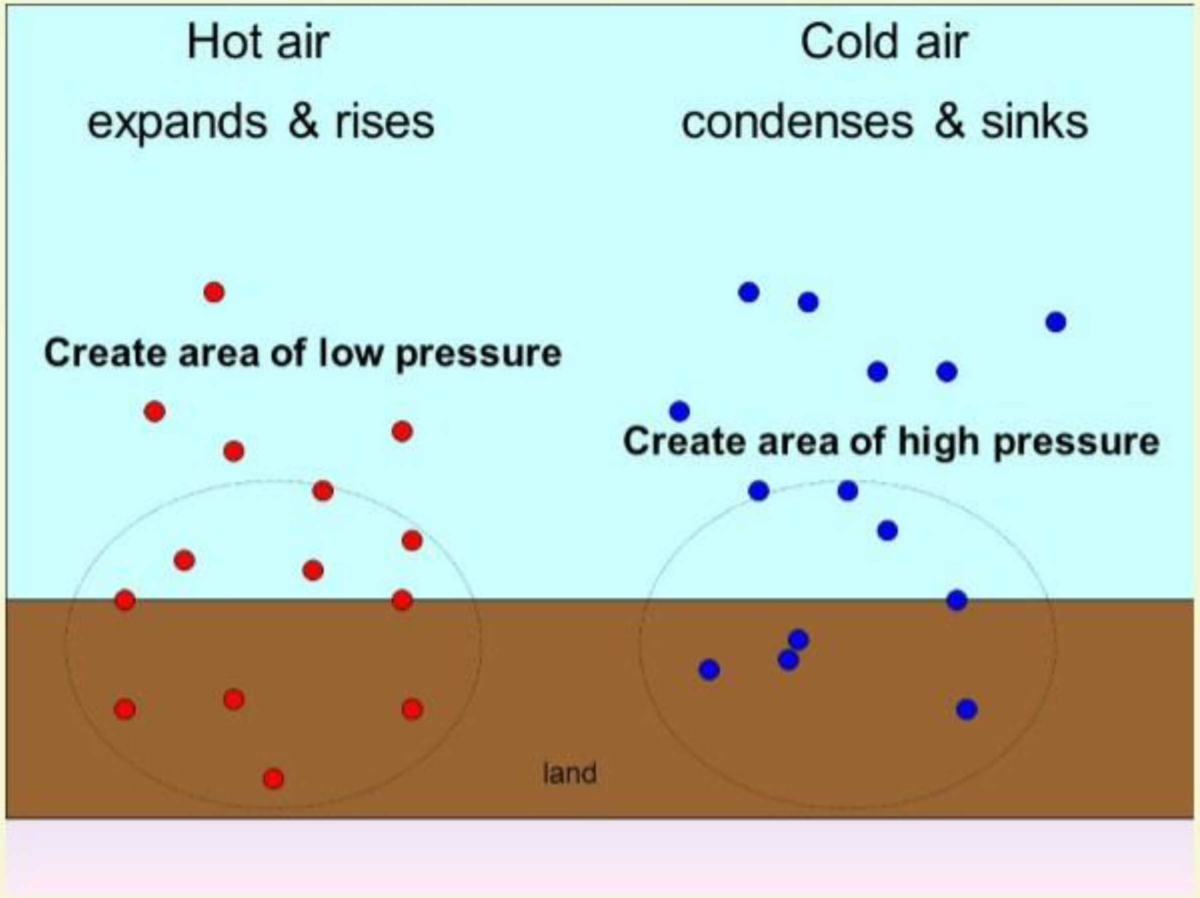
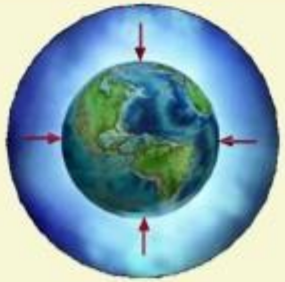




1. Air expands when heated and gets compressed when cooled. This results in variations in the atmospheric pressure.
2. Atmospheric pressure also determines when the air will rise or sink.



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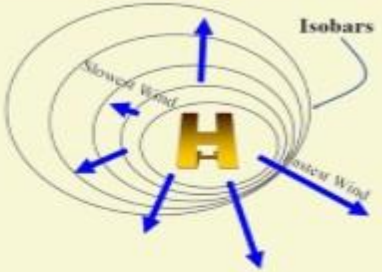
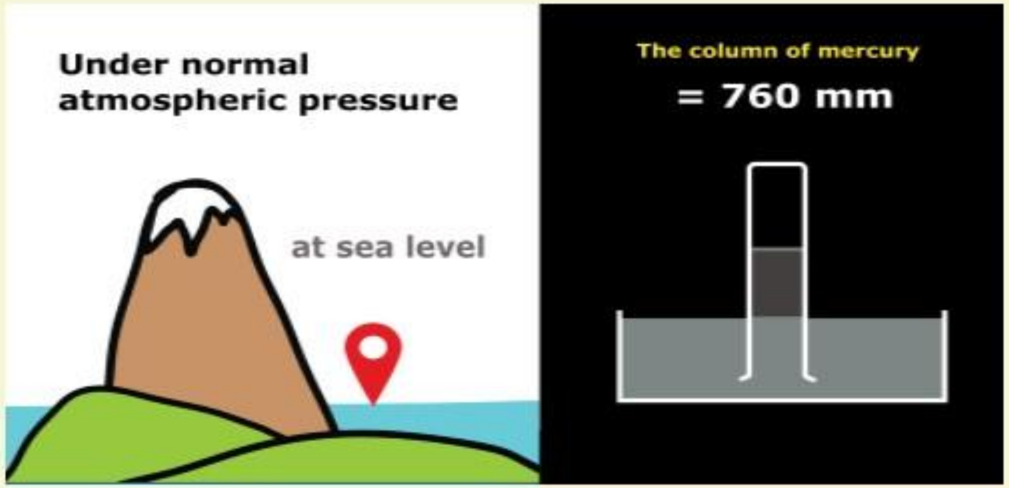


Pressure decreases with height

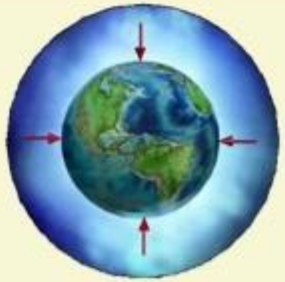
Level	*Pressure in mb*	*Temperature °C*
Sea Level	1,013.25	15.2
1 km	898.76	8.7
5 km	540.48	–17. 3
10 km	265.00	– 49.7



VARIATION OF PRESSURE WITH HEIGHT

1. At any elevation, pressure varies from place to place and its variation is the primary cause of air motion ie winds move from high pressure to low pressure areas.
2. In the upper atmosphere, pressure decreases rapidly with height.
3. However this decrease in rate of pressure is not uniform throughout and varies with altitude.

Learning Space



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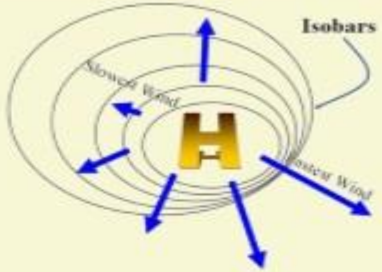
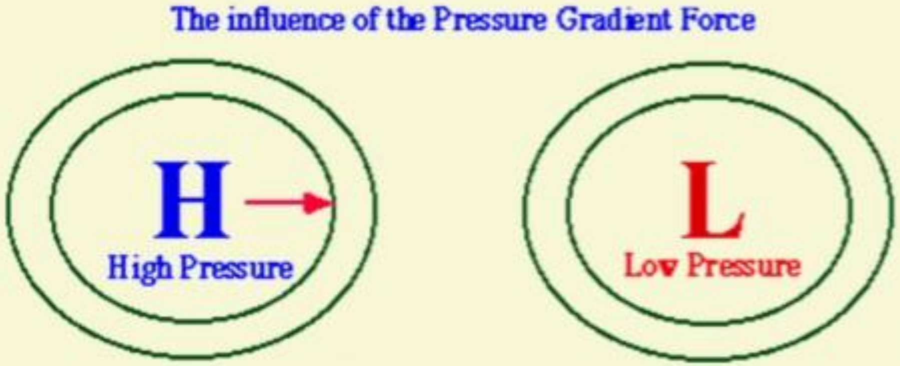
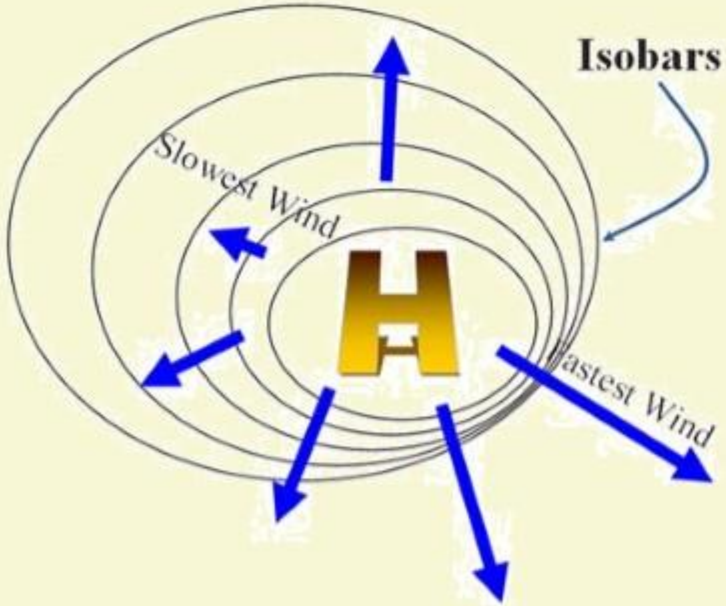




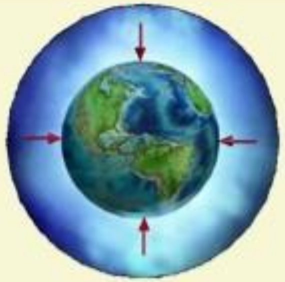


PRESSURE GRADIENT

1 The gradual change of pressure between different areas is known as the barometric slope or more commonly as the pressure gradient.

2 Widely spaced isobars => weak pressure gradient => slow winds; Closely spaced isobars => high pressure gradient => fast winds;

3 The pressure gradient results in a net force that is directed from high to low pressure and this force is called the "pressure gradient force".



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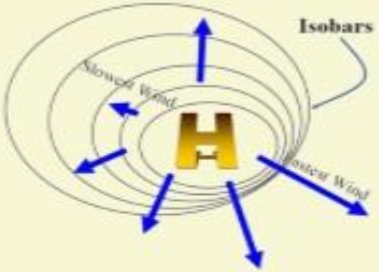
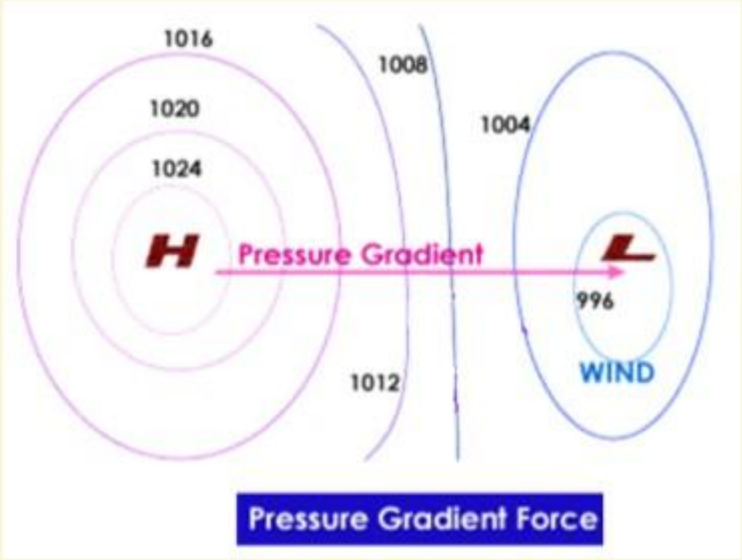
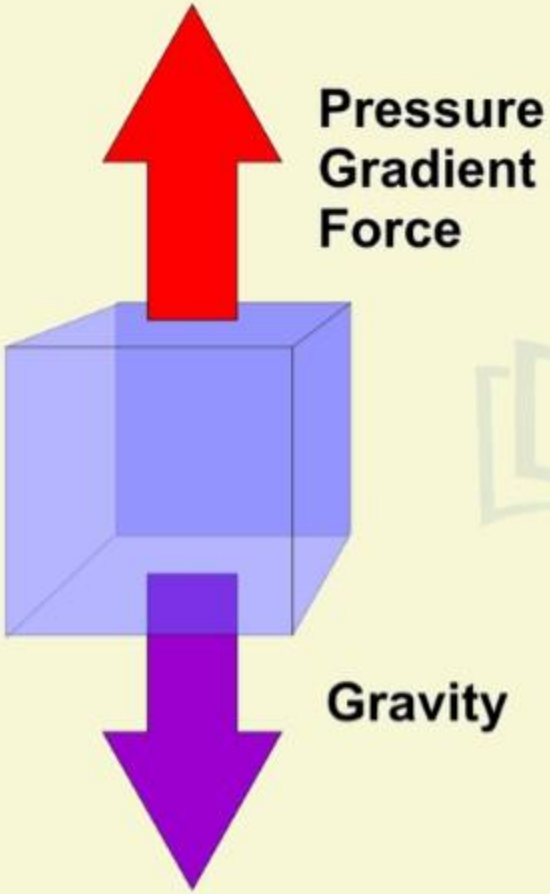
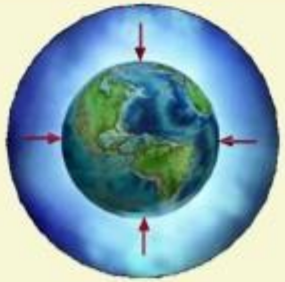






WHY NO STRONG VERTICAL WIND?

1. The vertical pressure gradient force(PGF) is much larger than that of the horizontal pressure gradient.
2. But, it is generally balanced by a nearly equal but opposite gravitational force. Hence, we do not experience strong upward winds.



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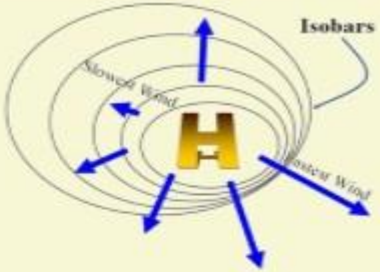
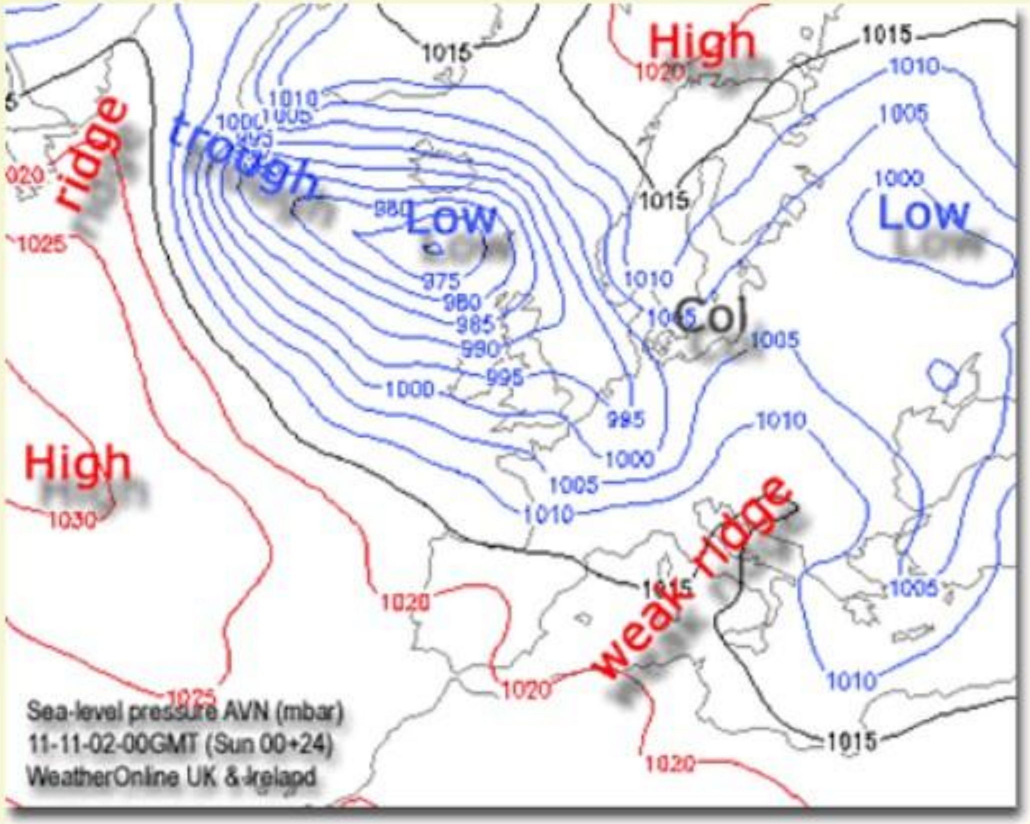
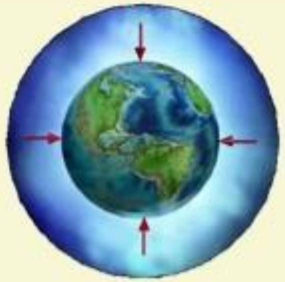


HORIZONTAL DISTRIBUTION OF PRESSURE



1. Small differences in pressure are highly significant in terms of the wind direction and velocity.
2. Horizontal distribution of pressure is studied by drawing isobars.
3. Isobars are lines connecting places having equal pressure reduced to sea level.
4. In order to eliminate the effect of altitude on pressure, it is measured at any station after being reduced to sea level for purposes of comparison.



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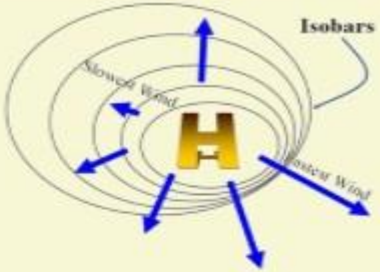
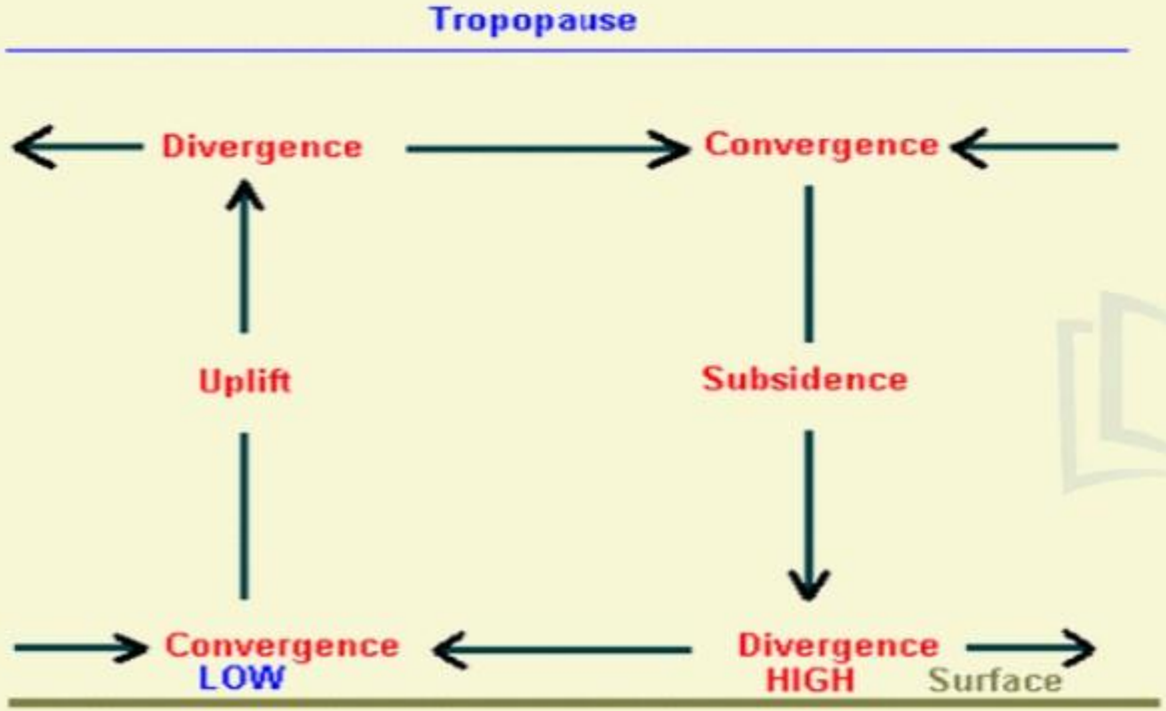


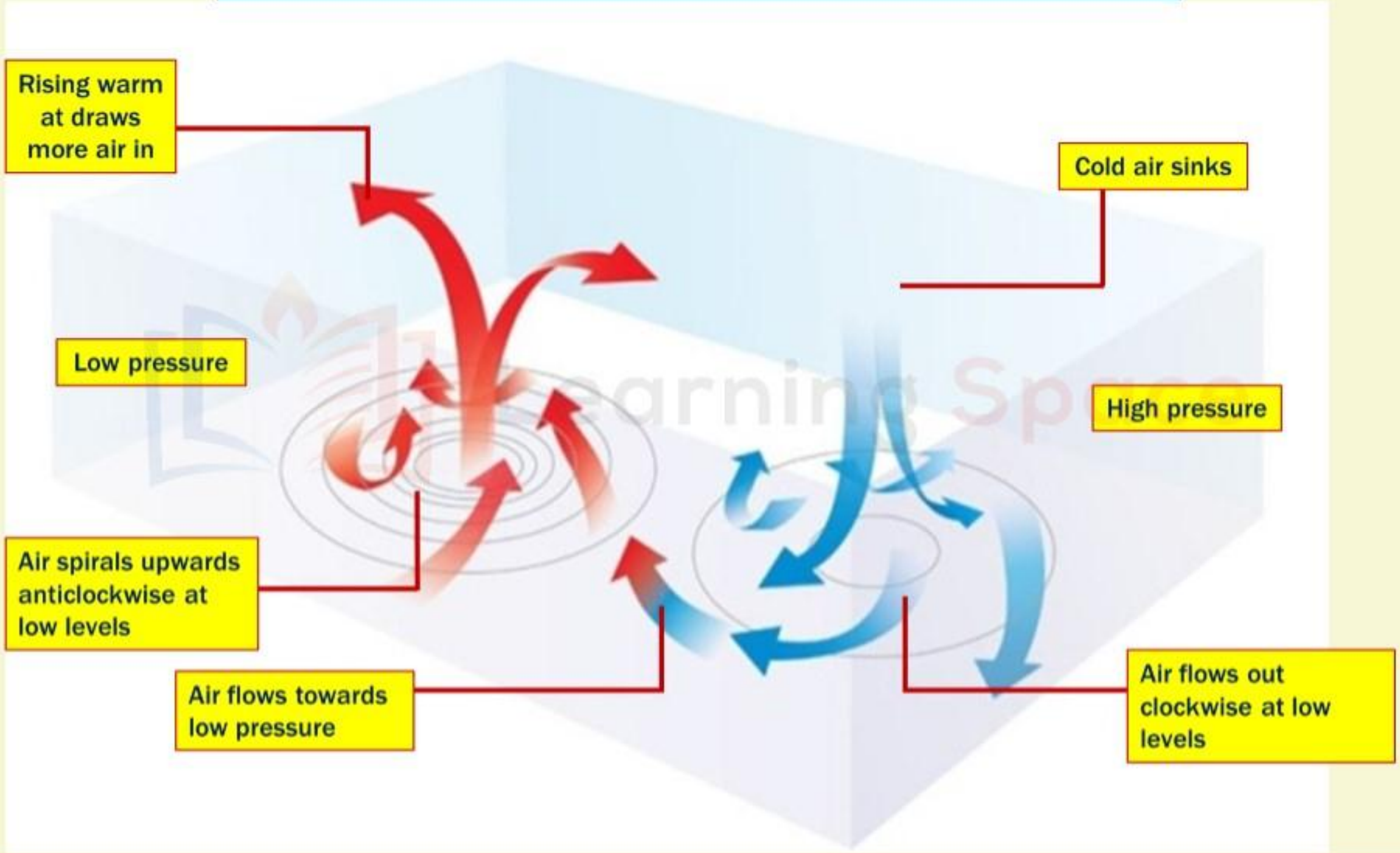
CONVERGENCE AND DIVERGENCE OF WINDS



1. The wind circulation at the earth's surface around low and high on many occasions is closely related to the wind circulation at higher level.
2. Generally, over low pressure area the air will converge and rise.
3. Over high pressure area the air will subside from above and diverge at the surface.

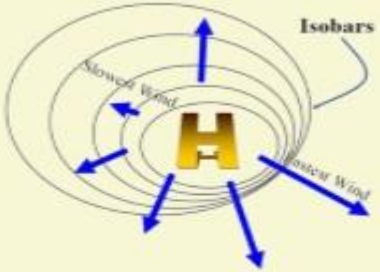
ATMOSPHERIC PRESSURE

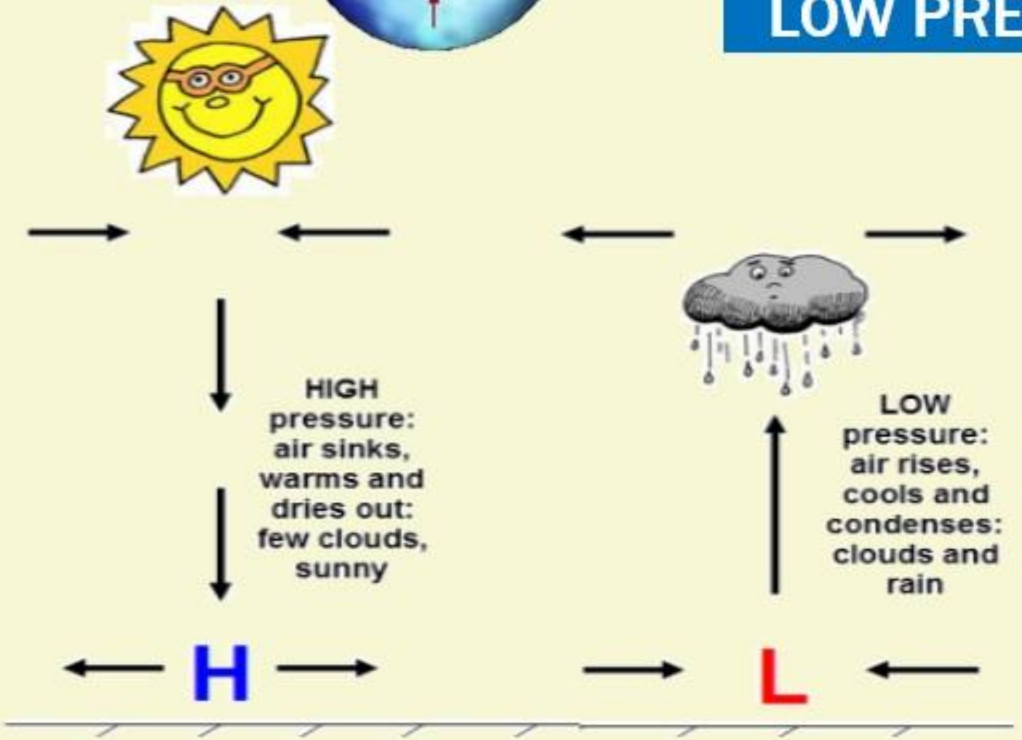
CONVERGENCE AND DIVERGENCE OF WINDS

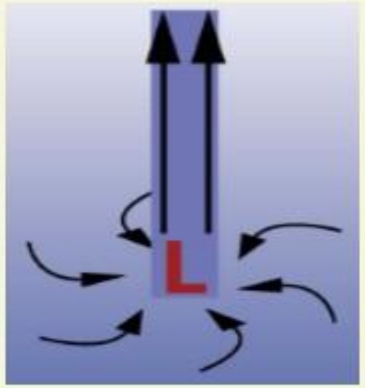


ATMOSPHERIC PRESSURE

LOW PRESSURE SYSTEM

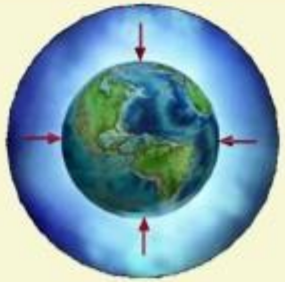






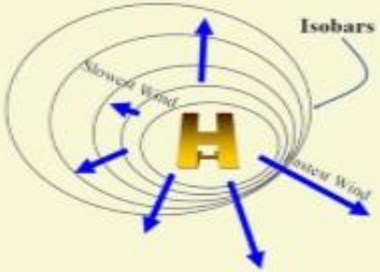
1. Low-pressure system is enclosed by one or more isobars with the lowest pressure in the centre.
2. A low pressure is also known as depression or a cyclone depending on its intensity.
3. Winds blow towards the low pressure, and the air rises upwards into the atmosphere where they meet
4. As the air rises, the water vapor within it condenses forming clouds and often leads to precipitation.
5. Hence, it is said that a low pressure system brings INSTABILITY (thunderstorms and rains) to the place.

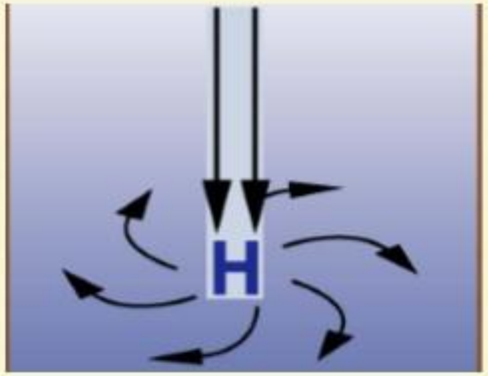
Learning Space

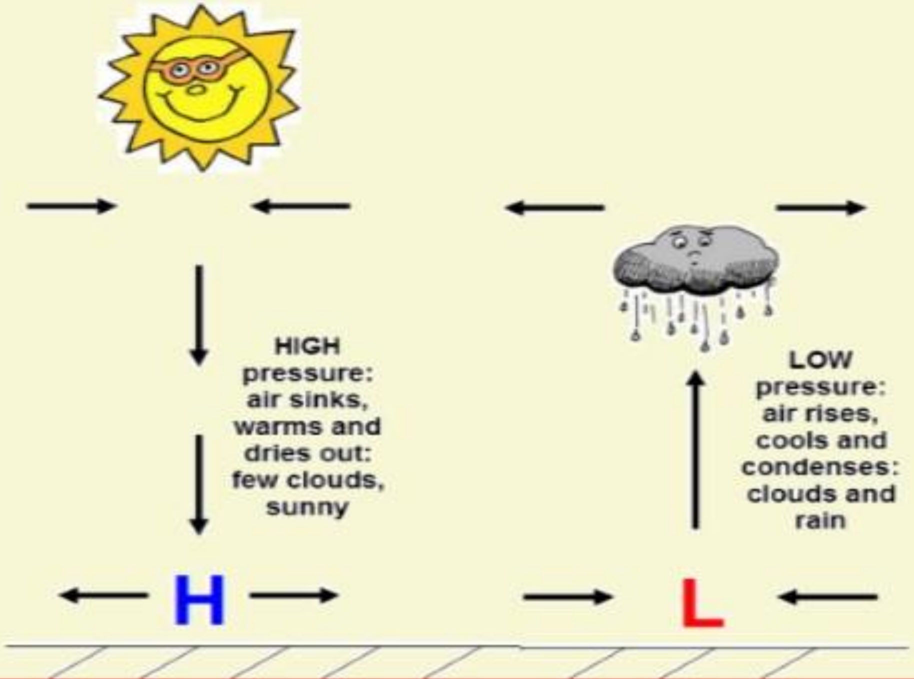


ATMOSPHERIC PRESSURE

HIGH PRESSURE SYSTEM

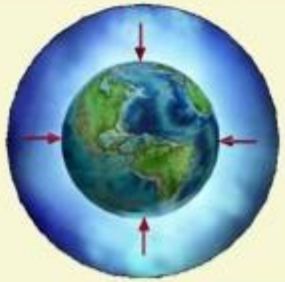






1. High-pressure system are comparatively larger system enclosed by one or more isobars with the highest pressure in the centre.
2. It is also known as anti-cyclone.
3. The convergence of winds at higher altitudes above this zone result in the subsidence of air.
4. Thus, descent of winds result in the contraction of their volume and ultimately causes high pressure.
5. Descent of an air parcel leads to rise of temperature. That is high pressure brings in sunny weather and hence it is said that HP brings ATMOSPHERIC STABILITY or in extreme cases ARIDITY.

Learning Space



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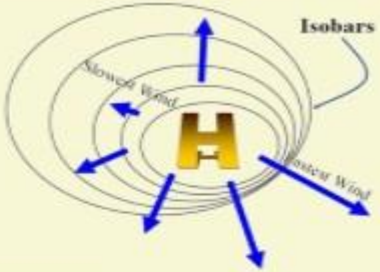
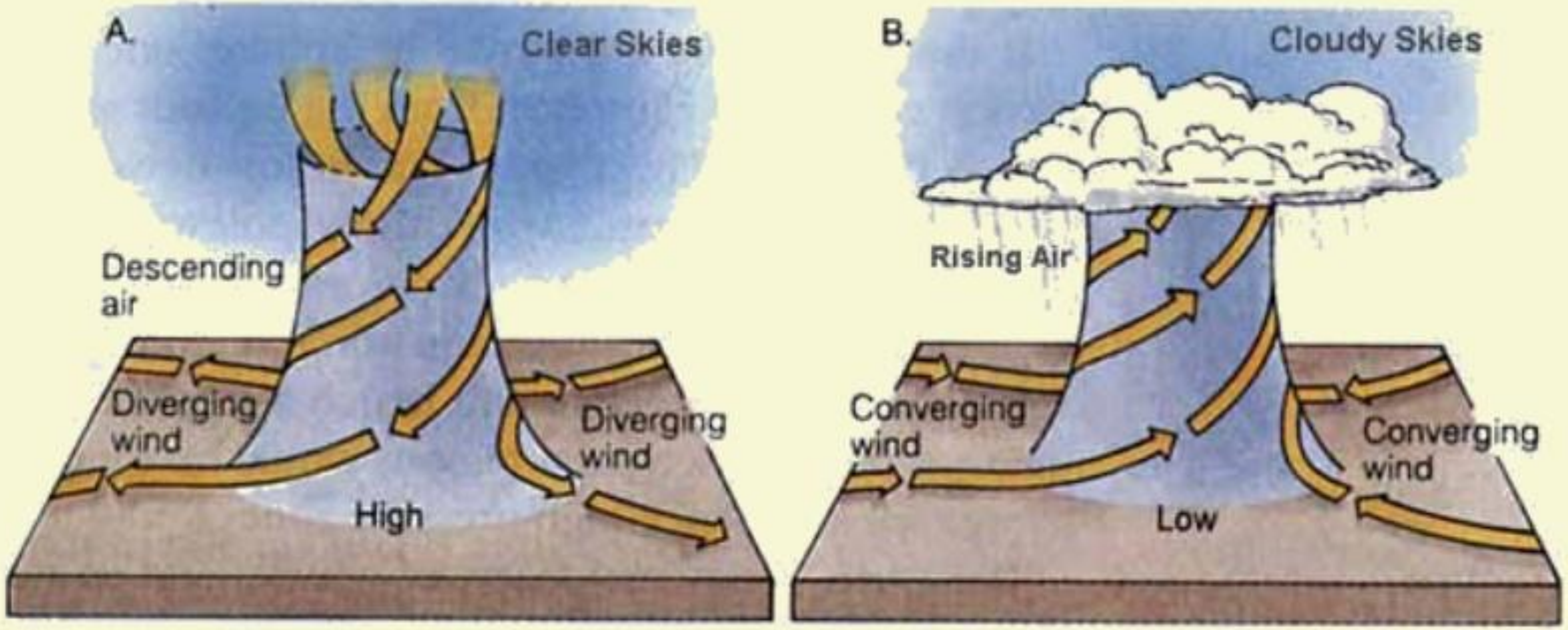
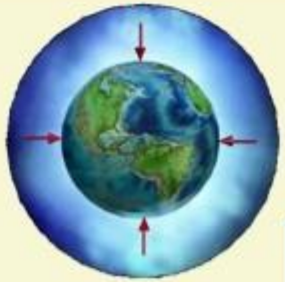


HIGH PRESSURE VS LOW PRESSURE SYSTEM



High Pressure brings stability/aridity

Low pressure brings instability/thunderstorms



ATMOSPHERIC PRESSURE

UPSC 2013 GS-1 QUESTION

Major hot deserts in northern hemisphere are located between 20-30 degree north and on the western side of the continents. Why? (10 Marks)

HINTS

1. Offshore trade winds in the region
2. Prevailing anticyclonic conditions- Areas between 20–30 degree latitudes on western margins of continents are the regions of descending air
3. Presence of cold ocean currents along the western coast of continents- this stabilizes the air and prevents cloud formation and rainfall.
4. Leeward sides of mountains/Parallel mountain ranges

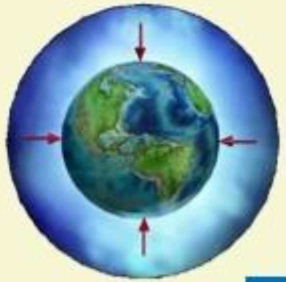

Barents Sea
Baffin Bay
ALASKA
CANADA
RUSSIA
Great Basin
Mojave
Sonoran
Chihuahuan
Desert
ED STATES
North Pacific Ocean
North Atlantic Ocean
Kara Kum
Lut
Thar
Deserts
Gobi
Taklamakan
Deserts
CHINA
North Pacific Ocean
Sahara Desert
Arabian
Desert
Somali
Chalbi
Desert
South Atlantic Ocean
Indian Ocean
BRAZIL
BOLIVIA
COLOMBIA
VENEZUELA
Peruvian
Atacama
Deserts
South Pacific Ocean
Namib
Kalahari
Deserts
Great Sandy
Simpson
Desert

MAJOR HOT DESERTS OF THE WORLD

LUT DESERT IN IRAN

*Lut Desert is a **UNESCO World Heritage Site** as it falls under the criteria of exceptional example of ongoing geological processes and is well known for the "yardangs"*

Learning Space



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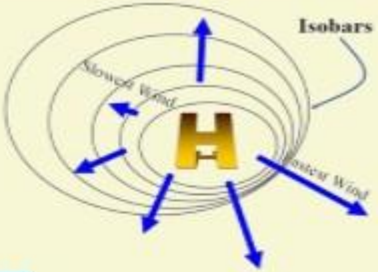
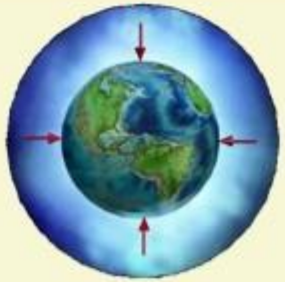


MAJOR HOT DESERTS OF THE WORLD

1. Great Sandy, Victoria, Simpson, Gibson and Sturt deserts in Australia
2. Kalahari Desert in south-western Africa
3. Sahara Desert in North Africa
4. Arabian Desert in Arabian peninsula
5. Thar Desert in India and Pakistan
6. Sonoran Desert in North and Central America
7. Mojave Desert in USA
8. Chihuahuan Desert in north central Mexico



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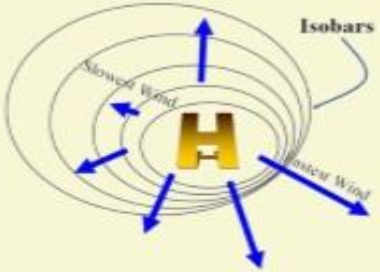


UPSC PRELIMS 2011

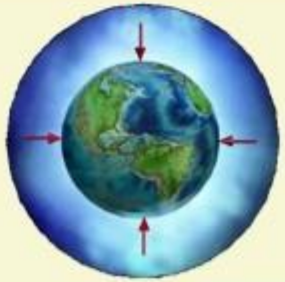
What could be the main reason/reasons for the formation of African and Eurasian desert belt?

1. It is located in the sub-tropical high pressure cells.
2. It is under the influence of warm ocean currents.

Which of the statements given above is/are correct in this context?

a) **1 only**
b) 2 only
c) Both 1 and 2
d) Neither 1 nor 2

Learning Space



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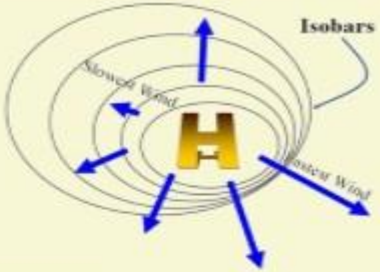

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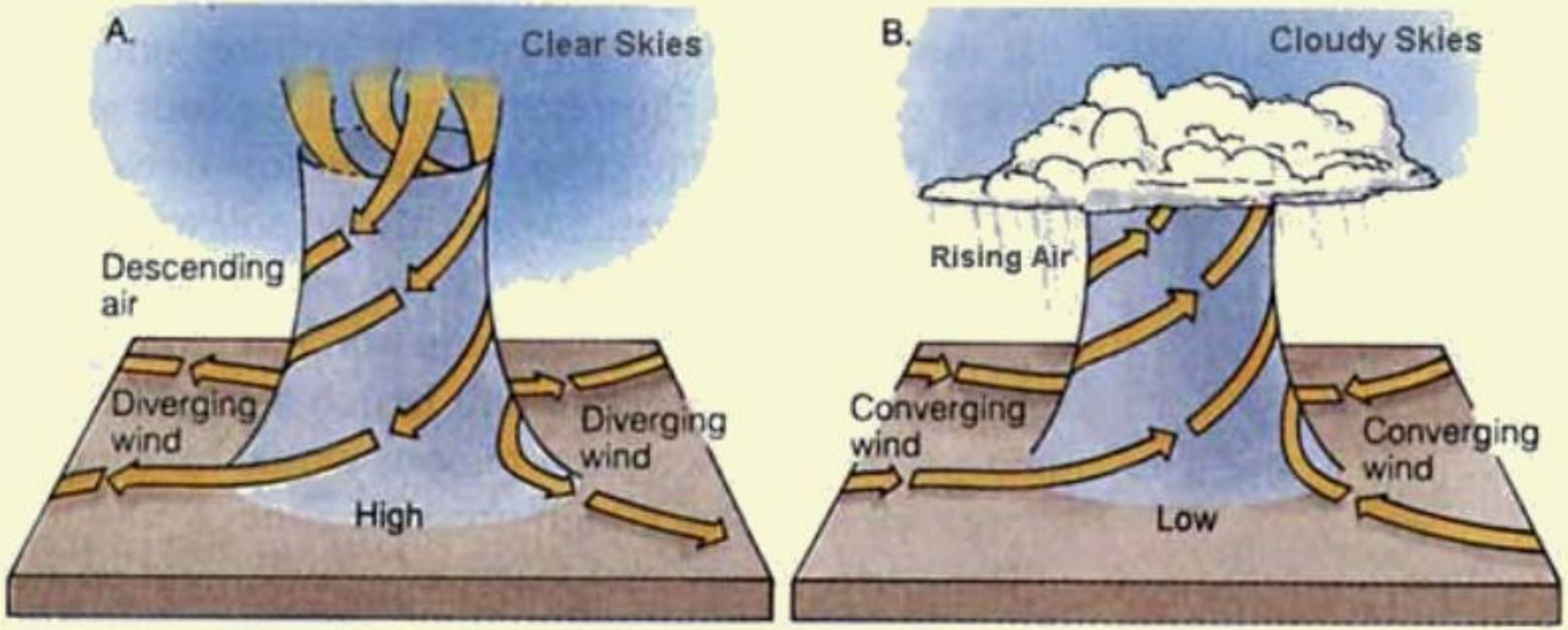

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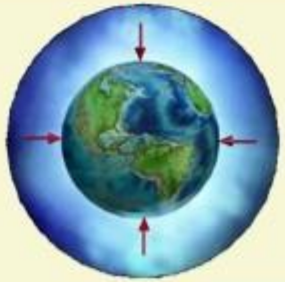

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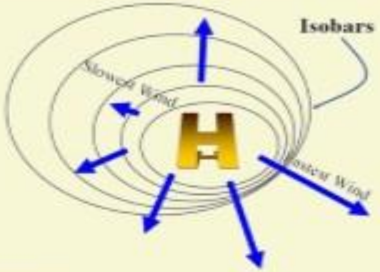
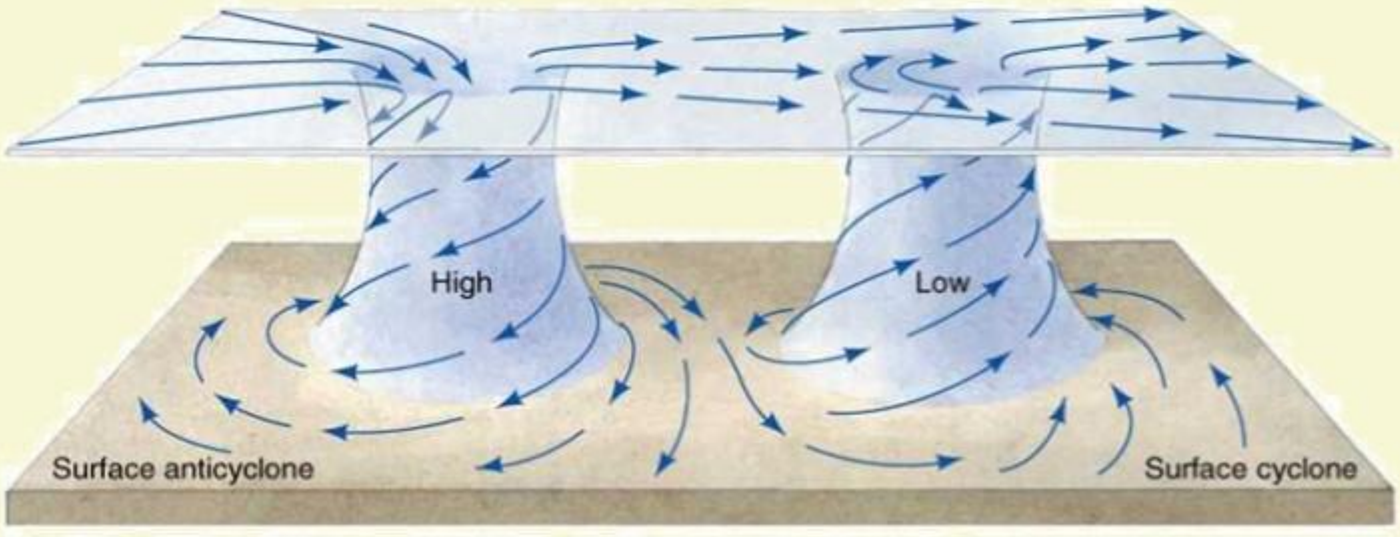
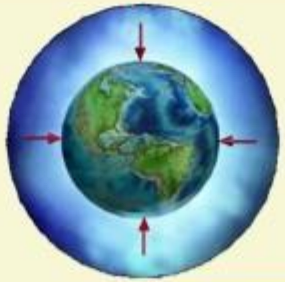


CYCLONIC/ANTI-CYCLONIC CIRCULATION



1. The wind circulation around a low is called as cyclonic circulation.
2. The wind circulation around a high is called anti-cyclonic circulation.

Learning Space



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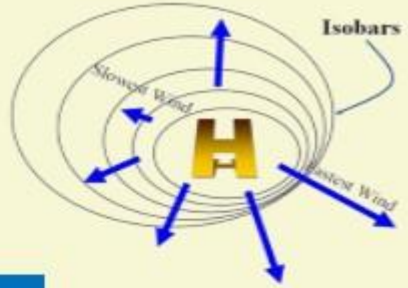
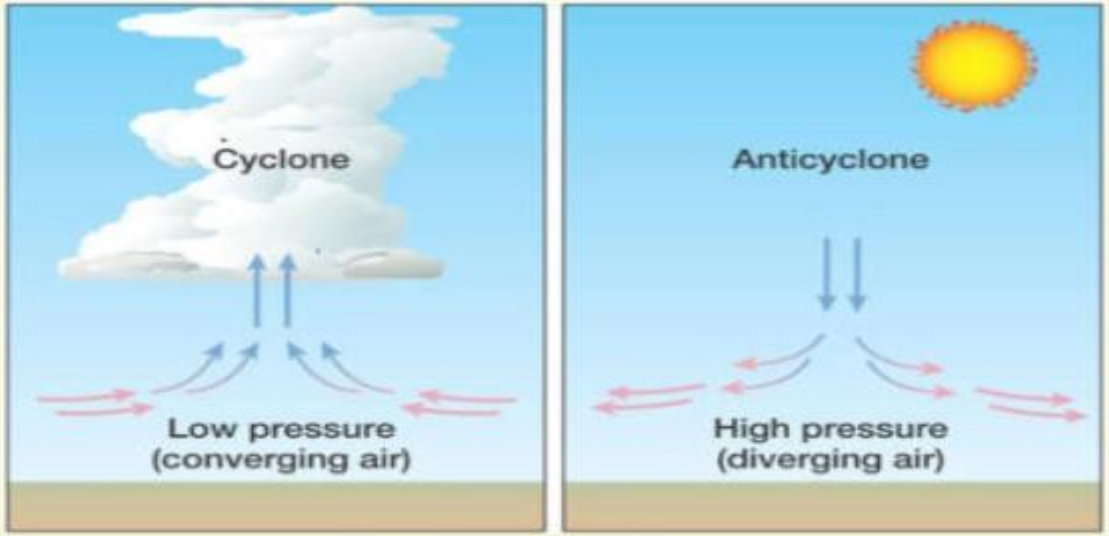


CYCLONIC/ANTI-CYCLONIC CIRCULATION



Pattern of Wind Direction in Cyclones and Anticyclones			
Pressure System	*Pressure Condition at the Centre*	*Pattern of Wind Direction*	
		Northern Hemisphere	*Southern Hemisphere*
Cyclone	Low	Anticlockwise	Clockwise
Anticyclone	High	Clockwise	Anticlockwise

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